JALA Academy
Assignment 4

Lists (Arrays) in Python

Python solution reference • Code examples are organized by the assignment questions.

How to use this document

Save the code examples into a .py file and run with Python 3. Where a program asks for input, enter values when prompted. For file-handling exercises, create the named text file in the same folder first.

1–10. List operations

```
def list_sum(items):
    return sum(items)

def list_average(items):
    return sum(items) / len(items) if items else 0

def find_index(items, target):
    try:
        return items.index(target)
    except ValueError:
        return -1

def contains(items, target):
    return target in items

def remove_element(items, target):
    result = items.copy()
    if target in result:
        result.remove(target)
    return result

def copy_list(items):
    return items.copy()

def insert_at(items, index, value):
    result = items.copy()
    result.insert(index, value)
    return result

def minimum_maximum(items):
    if not items:
        return None, None
    return min(items), max(items)

def reverse_list(items):
    return items[::-1]

def duplicates(items):
    seen, repeated = set(), set()
    for item in items:
        if item in seen:
            repeated.add(item)
        seen.add(item)
    return sorted(repeated)

values = [4, 2, 7, 2, 9, 4]
print("Sum:", list_sum(values))
print("Average:", list_average(values))
print("Index of 7:", find_index(values, 7))
print("Contains 8:", contains(values, 8))
print("Remove 2:", remove_element(values, 2))
print("Copy:", copy_list(values))
print("Insert:", insert_at(values, 2, 99))
print("Min and max:", minimum_maximum(values))
print("Reverse:", reverse_list(values))
print("Duplicates:", duplicates(values))
```